

Jayanth Manikanta Karri

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years designing, deploying, and scaling production AI systems, spanning **Agentic AI pipelines**, **LLM fine-tuning and quantization**, and **hybrid cloud / on-premise GPU inference**. Hands-on with the latest AI advancements: multi-agent orchestration frameworks (LangGraph, AutoGen, CrewAI), Retrieval-Augmented Generation, multimodal models, and MCP-integrated tool-use agents. Proven outcomes: **35% inference latency reduction**, **70% faster content pipelines**, and **60% reduction in manual audit effort**. Seeking an AI/ML or Agentic AI engineering role focused on building and scaling next-generation intelligent systems.

TECHNICAL SKILLS

- **Agentic AI:** LangGraph, LangChain, LlamaIndex, AutoGen, CrewAI, ReAct, Chain-of-Thought, Tool-Use Agents, MCP (Model Context Protocol), Multi-Agent Orchestration, Agentic RAG
- **AI / ML:** LLMs (LLaMA 3.1, Qwen, Mistral, GPT-4), Diffusion Models (SDXL, FLUX), Fine-tuning, Quantization (GPTQ / AWQ), Prompt Engineering, Computer Vision (YOLOv8), PyTorch, Huggingface, Transformers
- **Deployment:** Docker, GPU Inference Orchestration, CPU Offloading, Load Balancing, Multi-node Serving, Hybrid Cloud / On-Prem Deployment, FastAPI, Apache, SSL / HTTPS, Linux, REST APIs, Git/GitHub
- **Languages:** Python, JavaScript, Java, C++
- **Cloud:** AWS (EC2, S3, SageMaker), Azure (AI Services, VM), GCP (Vertex AI, Cloud Run)
- **Databases:** PostgreSQL, MongoDB, MySQL, FAISS, MongoDB Atlas Vector Search
- **Frontend:** React.js, Next.js, TailwindCSS, Node.js, Express.js

EXPERIENCE

AI / ML Engineer — Agentic AI & Deployment Infrastructure

Jul 2025 – Present

Quantum AI Global | Hyderabad, Telangana

- Designed and operated a **hybrid AI deployment architecture** serving concurrent LLM and Diffusion Model workloads across **on-premise GPU servers** (Ada RTX 6000, 48 GB VRAM) and **cloud environments** (AWS EC2 GPU, Azure AI), enabling seamless failover and cost-adaptive routing between environments.
- Architected and shipped **multi-agent AI pipelines** using **LangGraph** and **AutoGen**, enabling autonomous task decomposition, tool-use, memory management, and self-correcting reasoning loops across production workflows.
- Integrated **Model Context Protocol (MCP)** for dynamic tool and data-source connectivity in agentic systems, reducing the time to connect new capabilities to agents from days to hours.
- Built parallel **FastAPI inference pipelines** with CPU offloading and multi-threaded GPU scheduling supporting simultaneous serving of LLaMA 3.1, Qwen, Mistral, SDXL, and FLUX with no cold-start downtime.
- Containerized AI services with **Docker** and orchestrated multi-node deployments linking frontend, inference engine, and backend nodes through secured internal networking and **cloud VPC peering**.
- Conducted **IOPS, network latency, and GPU utilization profiling** across on-prem and cloud endpoints; established performance baselines now used across all production AI deployments.

Junior AI Developer

Nov 2024 – Jul 2025

Quantum AI Global | Hyderabad, Telangana

- Built **agentic RAG systems** using **LangChain** and **LlamaIndex** implemented ReAct-style agents with tool-use, multi-hop retrieval, and self-reflection loops for enterprise document Q&A.
- Deployed and served open-source **LLMs (LLaMA 3.1, Qwen, Mistral)** and **Diffusion Models (SDXL, FLUX)** in production managed model loading, request batching, and API exposure via FastAPI backends.
- Applied **GPTQ and AWQ quantization** with VRAM optimization techniques cut inference latency by **35%** in shared-GPU environments with no measurable accuracy loss.
- Worked with **Vision-Language Models (VLMs)** including LLaVA and GPT-4V for multimodal pipelines combining image understanding with structured text generation.
- Built production computer-vision features **object detection, segmentation, and classification** using **YOLOv8** and **ViT**-based architectures integrated into the FastAPI backend.

- Contributed to the internal **MRC (Models, Routes, Controllers)** AI backend framework, enforcing modular separation that reduced new-model onboarding time and improved maintainability across **two shipped products**.

ML Engineer Intern

Mar 2024 – Nov 2024

Quantum AI Global | Hyderabad, Telangana

- Built **RAG pipelines** using **FAISS** and **MongoDB Atlas** vector search improved retrieval precision by **20%** through hybrid dense-sparse indexing for enterprise chatbot PoCs.
- Developed **FastAPI** AI API layers integrating **GPT-4** and **Claude** APIs into intelligent automation workflows delivered **4 PoC prototypes** adopted into the production roadmap.
- Explored **structured outputs, function calling, and JSON mode** across OpenAI and Anthropic APIs to build reliable, schema-constrained AI responses for downstream application integration.
- Researched and documented **prompt engineering, chain-of-thought reasoning, and chatbot personalization** strategies outputs became team-wide prompt design standards.

PROJECTS

Autonomous AI Marketing Agent

- Built a **multi-agent system** using **CrewAI** where specialized agents autonomously handle brief analysis, visual generation (SDXL, FLUX), copywriting (LLaMA, GPT), and campaign scheduling reducing human touchpoints per campaign by **80%**.
- Deployed across **AWS** (burst inference) and **on-prem GPU servers** (steady-state) with a **Next.js** frontend and MongoDB campaign storage cut total content creation time by **70%**.

Intelligent Policy Analyzer & Risk Reporting System (IPARRS)

- Enterprise AI compliance system powered by an **agentic RAG pipeline** (LangChain + FAISS + PostgreSQL) agents autonomously retrieve policy clauses, cross-reference regulatory rules, and generate risk reports via a **React.js** dashboard.
- Reduced manual audit effort by **60%** **semantic re-ranking** and multi-hop agent reasoning deliver high-precision risk flagging across complex policy documents.

LEADERSHIP

Product Lead — Quantum AI Global

- Led development of **two AI-based entertainment products** from zero to deployment managed a team of **5 developers and 3 interns** across feature planning, sprint execution, and agentic AI integration.
- Mentored team members in **LLM integration, agentic workflow design, and backend API development**; introduced code-review practices that improved PR throughput by **40%**.

EDUCATION

B.Tech — Computer Science, Specialization in AI & ML

2021 – 2025

Gayatri Vidya Parishad College of Engineering (Autonomous) | CGPA: 8.8 / 10

Full-Stack & MERN Development — NxtWave Genius Track Certification

Intensive programme by NxtWave (IIT alumni-backed); built production-grade apps using MongoDB, Express.js, React.js, and Node.js.

ACHIEVEMENTS & PUBLICATIONS

- **Winner, Smart India Hackathon 2023** — National-level hackathon; built an AI multilingual video dubbing system for educational content combining NLP + TTS + ML (team of 6). [View post](#)
- **Publication:** *Remote Healthcare Diabetic Retinopathy Detection Using Deep Learning* — STM Journals, 2023. [Read here](#)